ANISH R. VERMA | RÉSUMÉ

EDUCATION

MSc - Theoretical Nuclear Physics — University of Guelph, 2019 - Present

BSc Honours - Chemical Physics — Simon Fraser University, 2013 - 2018

- Faculty of Science Graduating Class of 2018 Speaker (**Video**)

EXPERIENCE

Quantitative Analyst / Quantum Software Developer — 1QBit, 2020 - Present

- **Mathematically modeled** risk for futures, **analyzed** major price moves, and produced a series of **white papers** and **technical reports**.
- **Developing** the **quantum** version of the **Market Sentiment Meter** that makes use of **quantum optimization algorithms**.
- **Researched** efficacy of novel mixture distribution for risk in **trading algorithm**, with results to be published in a forthcoming paper.

Chief Data Officer — STEM Fellowship, 2020 - Present

- **Led** several large teams in organizing national big data **challenges**, to empower students to use data science in an interdisciplinary setting.
- **Designed** workshops with topics ranging from **statistics** to **machine learning**.
- **Directed** data analytics nationally, to **quantify** and improve internal operations.

Research Assistant — Guelph-Waterloo Physics Institute, 2019 - Present
Supervised by: Prof. Alexandros Gezerlis

- **Mathematically modeled** quasiparticle excitations in nuclear matter by extending finite Fermi liquid theory to the thermodynamic limit for strongly interacting neutrons.
- Performed highly precise and demonstrably accurate **Quantum Monte Carlo Simulations** on **cluster supercomputers**, housed by Compute Canada.
- Funded by the Natural Sciences and Engineering Research Council of Canada (**NSERC**) through the Canada Graduate Scholarship Master's (**CGS-M**) Program.

Research Assistant — Simon Fraser University, 2016 - 2017
Supervised by: Prof. Krzysztof Starosta

- **Mathematically modeled** pair-breaking in quadrupole deformed nuclei and **predicted** its effects on observables by treating both single-particle and collective degrees of freedom.
- Model **reduced the computational time** required to diagonalize the large matrices usually involved in nuclear structure calculations, by significant reduction of the phase space.
- Funded by **NSERC** and the Canadian Institute of Nuclear Physics (**CINP**), resulting in **publication** and several **conference presentations**; Awarded 2nd place **presentation award** in the division of nuclear physics at a national conference (CAP Congress 2017).

CONTACT

604-446-8988
anishrverma@gmail.com
anishrverma.github.io
/in/AnishRVerma/

GENERAL SKILLS

Data Analysis ★★★★☆
Mathematical Modeling ★★★★☆
Presentation and Communication ★★★★★
Project Management ★★★★★

LANGUAGES

Python
MATLAB
Mathematica
Bash
Fortran
Git

SOFTWARE

LaTeX, Excel, Word, PowerPoint, Inkscape, Photoshop, InDesign, LabVIEW, Igor Pro

OPERATING SYSTEMS

INTERESTS

References available upon request.